

Lesson 16 · Investment flows and accumulated decisions

10 single-choice questions · 20 minutes · 3 Foundation + 3 Core + 3 Application + 1 optional Challenge. Choose one answer per question.
Work at your own pace; partial completion is welcome.

Q1 · Rank 1 / Foundation

If the investment flow of a company is $I(t) = 8000t^{1/3}$, calculate the capital formation during the first eight years.

☐ A 96 000☐ B 128 000☐ C 76 800☐ D 256 000

Jacques, 9th ed. · §6.2.3 · Chapter 6 Multiple Choice Questions Question 5 · printed p. 516 / PDF p. 533

Q2 · Rank 1 / Foundation

Given investment flow $I(t) = 2400\sqrt{t}$, calculate total capital formation during the first four years.

☐ A 19 200☐ B 12 800☐ C 6400☐ D 4800

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 9(a) · printed p. 510 / PDF p. 527

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Q3 · Rank 1 / Foundation

For a green spaces project, investment is $I(t) = 0.6t^2 + 10$ million dollars per year for the first five years, then $J(t) = 25e^{0.4(5-t)}$ for $5 \leq t \leq 10$. Verify $I(5) = J(5)$. What is their common value?

☐ A 10 million dollars per year.

☐ B 75 million dollars.

☐ C 25 million dollars per year.

☐ D 25 million dollars.

Jacques, 9th ed. · §6.2.3 · Exercise 6.2* 13(a), boundary verification · printed p. 512 / PDF p. 529

Q4 · Rank 2 / Core

If investment flow is $I(t) = 5000t^{1/4}$, calculate capital formation from the end of the second year to halfway through the fifth year. Give the answer to the nearest whole number.

☐ A 20 000

☐ B 20 393

☐ C 16 703

☐ D 9500

Jacques, 9th ed. · §6.2.3 · Exercise 6.2 8 · printed p. 510 / PDF p. 527

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Q5 · Rank 2 / Core

Find an expression for capital formation between $t = 0$ and $t = T$ if net investment is $I(t) = At^a$, where A and a are positive constants.

☐ A AT^a

☐ B $AT^{a+1}/(a+1)$

☐ C AT^{a+1}

☐ D AaT^{a-1}

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Q6 · Rank 2 / Core

Find an expression for capital formation between $t = 0$ and $t = T$ if net investment is $I(t) = Ae^{at}$, where A and a are positive constants.

☐ A Ae^{aT}

☐ B $A(e^{aT} - 1)$

☐ C Ae^{aT}/a

☐ D $A(e^{aT} - 1)/a$

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Q7 · Rank 3 / Application

An oil company starts extraction at $t = 0$. The extraction rate, in thousands of barrels per year, is $dN/dt = 30t^2 - 4t^3$. Calculate the number of barrels extracted during the first four years.

☐ A 384 barrels.

☐ B 384,000 barrels.

☐ C 96,000 barrels.

☐ D 192,000 barrels.

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Q8 · Rank 3 / Application

Given investment flow $I(t) = 2400\sqrt{t}$, find an expression for annual capital formation during the N th year and hence the first year in which annual capital formation exceeds USD 4000. Choose the first year.

☐ A Year 1.

☐ B Year 3.

☐ C Year 4.

☐ D Year 5.

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Q9 · Rank 3 / Application

If net investment is $I(t) = 100e^{0.1t}$, calculate capital formation from the end of the second year to the end of the fifth year. Round to two decimal places.

☐ A 648.72☐ B 427.32☐ C 221.40☐ D 164.87

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Q10 · Rank 4 / Challenge · OPTIONAL

Calculate the present value of a continuous revenue stream of USD 1000 per year at a 5% discount rate, paid for 3 years. Use continuous discounting and round to two decimal places.

☐ A 3000.00☐ B 2591.51☐ C 2785.84☐ D 3153.42

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